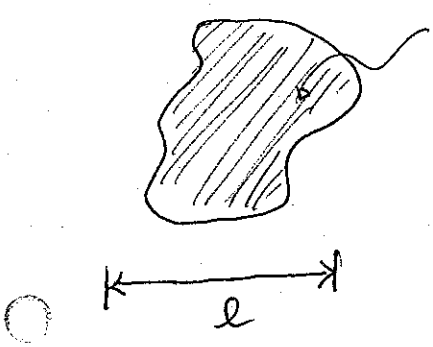
FLUIDS

The Properties of fluids can be derived from the molecular dynamics of a dilute gas for which the mean free path is small compared to the typical length scale of the system. Here we will derive the fluid properties from macroscopic considerations and treating the fluid as a continuum.



ℓ is chosen so that

$$\lambda_{mfp} \ll \ell \ll L$$

λ_{mfp}: mean free path for collisions

L: scale for large scale changes

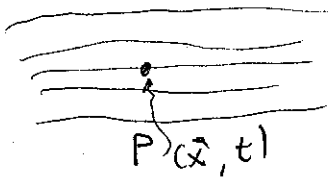
The system variables are $\rho(\vec{x},t)$, $\vec{v}(\vec{x},t)$, $p(\vec{x},t)$, $T(\vec{x},t)$, where, however; $\rho(\vec{x},t)$ and $p(\vec{x},t)$ are not independent since they are related through the Equation of state. This equation will be needed, in addition to the fluid equations that we'll derive, in order to have a closed set of equations. Examples of equations of state are:

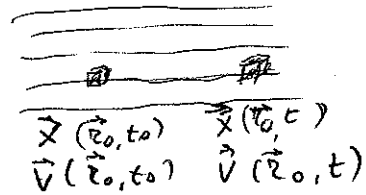
- $p = \rho k T/\mu$ (ideal gas)
- $p = A\rho^{\gamma}$ (polytropic; adiabatic)
- $p = \rho c_s^2$ (isothermal, where $c_s^2 = \frac{kT}{\mu}$ is the sound speed of the medium).

There can be two approaches to study fluids:

Eulerian method:
- Tracks flow through space
- No identity of fluid
- Concentrates on fluid properties at a point $P(x, y, z, t)$

Lagrangian method:
- Tracks fluid elements
- Identify fluid element by initial position $\vec{r}_0$ at t_0 and follow it with time





The Lagrangian description tends to be generally more cumbersome to use, but there are situations where one is specifically interested in following up the trajectories of specific elements of fluids with time. In this case the Lagrangian formulation would turn out to be handy. An example is in the study of supernova explosions where one is also interested in studying where the various metals inside the star end up with the explosion. In general, the Eulerian approach is simpler to use: it can be considered as a mean field approach. Now, given a field variable $f(\vec{x}, t)$, how does it vary with time because of fluid motion?

Let v_x, v_y, v_z be the component of the flow velocity.

Then $(x, y, z) \xrightarrow[\text{time } \delta t]{\text{after a}} (x + v_x \delta t,\ y + v_y \delta t,\ z + v_z \delta t)$

$$f(x, y, z, t) \xrightarrow[\text{time } \delta t]{\text{after a}} f(x + v_x \delta t,\ y + v_y \delta t,\ z + v_z \delta t,\ t + \delta t) \equiv f'$$

Making a Taylor expansion and neglecting terms above the first order in δt, we have:

$$f' = f(x, y, z, t) + V_x \delta t \frac{\partial f}{\partial x} + V_y \delta t \frac{\partial f}{\partial y} + V_z \delta t \frac{\partial f}{\partial z} + \delta t \frac{\partial f}{\partial t}$$

Let's now define the advective derivative (or also called Lagrangian derivative):

$$\frac{df}{dt} = \lim_{\delta t \to 0} \frac{f(\vec{x} + \vec{v}\delta t, t + \delta t) - f(\vec{x}, t)}{\delta t}$$

Then using the expansion above we obtain:

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \underbrace{V_x \frac{\partial f}{\partial x} + V_y \frac{\partial f}{\partial y} + V_z \frac{\partial f}{\partial z}}_{(\vec{v} \cdot \vec{\nabla}) f}$$

So we have the general expression:

$$\boxed{\frac{df}{dt} = \frac{\partial f}{\partial t} + (\vec{v} \cdot \vec{\nabla}) f}$$

In the case where the variable f is the velocity $\vec{v}$ of a particle, then one obtains: $\frac{d\vec{v}}{dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla})\vec{v}$

$\uparrow$ acceleration of particle

$$a_i = \frac{\partial V_i}{\partial t} + \sum_j \underbrace{\frac{\partial x_j}{\partial t}}_{V_j} \frac{\partial V_i}{\partial x_j}$$

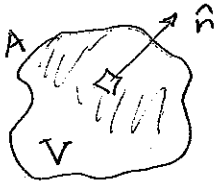
Note that: there can be acceleration even if $\frac{\partial V}{\partial t} = 0$ so that the velocity at a given point is not changing. An example is that of water flowing in a circle at a constant speed. This is accelerating even though the velocity at a given point is not changing.

EQUATIONS OF FLUID DYNAMICS
IDEAL FLUIDS

The equations of fluid dynamics are simply the equations of conservation of mass, momentum (eqn. of motion for) and energy (conservation of), modulo some algebraic manipulation

1. Conservation of Mass

Let's consider a fluid element of volume V and area A in which there are no sources or sinks



. The mass $M = \int_V \rho \, dV$ within that volume will vary only because of fluid motion (in or out of volume) if there are no sources or sinks:

$\frac{dM}{dt} = \int_V \frac{\partial \rho}{\partial t} dV$. Let's now determine the amount of mass that flows out of the area A in the interval dt. [This is a simple/intuitive demonstration of a formal result known as "Reynolds Transport Theorem"]

Let's consider a small area dA on the surface. The amount of mass that flows through an element dx within the time dt is given by $\frac{dm}{dt} = -\rho \, dA \frac{dx}{dt} = \rho v dA$.
(if $\vec{v}$ in direction of $+\hat{n}$, mass is decreasing $\rightarrow$ − sign)

If the motion of the flow is not perpendicular to the surface, one only has to consider the + comp. for motion of fluid in & out of surface.

In this case: $\frac{dm}{dt} = -\rho \, \vec{v} \cdot \underbrace{\hat{n} \, dS}_{d\vec{A}}$. The flow of mass throughout the all surface is then: $\frac{dM}{dt} = -\oint_A \rho \vec{v} \cdot \underbrace{\hat{n} \, dS}_{d\vec{A}}$

We can therefore write $\int_V \frac{\partial \rho}{\partial t} dV = -\oint_A \rho \vec{v} \cdot d\vec{A} \underset{\uparrow}{=} -\int_V \vec{\nabla} \cdot [\rho \vec{v}] dV$

[by Gauss' theorem $\oint_A \vec{Q} \cdot d\vec{A} = \int_V (\vec{\nabla} \cdot \vec{Q}) dV$]

Since this must be true for any V: $\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = 0$.

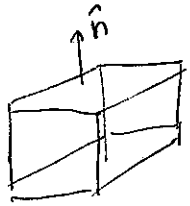
This can also be written as $\frac{\partial \rho}{\partial t} + \rho (\vec{\nabla} \cdot \vec{v}) + (\vec{v} \cdot \vec{\nabla}) \rho = 0$

$$\rightarrow \boxed{\frac{d\rho}{dt} + \rho \vec{\nabla} \cdot \vec{v} = 0}$$

CONTINUITY EQUATION

2. Momentum Equation (equation of motion)

This equation follows from Newton's second law of motion : $\vec{F}_{TOT} = M\vec{a}$



given an element of fluid, there can be two types of forces acting on it.

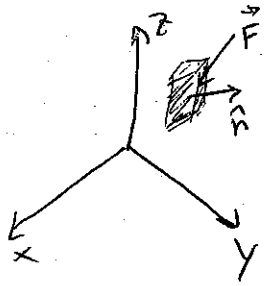
Forces on a fluid
- → Body forces: act at all points within the body of a fluid (e.g. gravity)
- → Surface forces: act on the fluid across the surface bounding the fluid element (e.g. pressure)

Recalling that $\vec{a} = \frac{d\vec{v}}{dt}$, the i component of the equation of momentum for a fluid of volume can be written as:

because forces are defined as acting on the surface

$$\int_V \frac{dv_i}{dt}\rho\, dV = \int_V \rho g_i\, dV - \oint_A P_{ij}\, dS_j$$

g_i: force per unit mass; P_{ij}: force per unit surface



In general, given a surface element, one can have a force in any given direction x, y, z, acting on a surface with any orientation x, y, z.

In order to account for all these possibilities, the force per unit area must be a second rank tensor, P_{ij}, defined so that P_{ij} = stress in i direction due to strain in j direction.

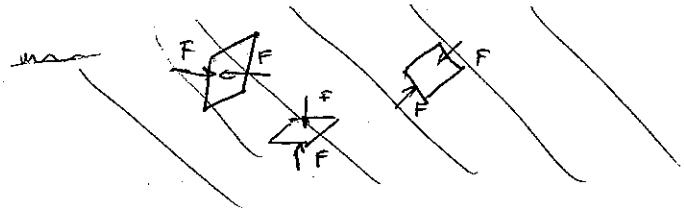
P_{ij}: stress tensor.

The surface force per unit Area is called stress.

The total stress in a given direction is given by $\vec{P}_i = P_{ij} \cdot \hat{n}_j$: If $\vec{P} \parallel \vec{n} \Rightarrow$ tensile stress

$P \perp \vec{n} \Rightarrow$ shear stress.

If there are no body torques: $P_{ij} = P_{ji}$



For a static fluid, it is an experimentally established fact that the force acting across an element of area inside the fluid or on its boundary is always perpendicular to that element of area. In fact, as we discussed, this is generally taken as the definition of a fluid (a fluid is defined as a substance in which motions are induced whenever there is a part of the surface force not $\perp$ to surface (i.e. a shear force), - in constrast with solids within which shear forces can be present in static equilibrium). Moreover, the force per unit area is the same for all orientations of the surface. Mathematically, for a static fluid:

$P_{ij} = p\,\delta_{ij}$. This equation doesn't generally hold when the fluid is in motion. In this case, shear forces induced by viscosity are always present. However, in many situations where viscosity does not play an important role, one can use this assumption, which results in much simpler equations to solve (these fluids are known as "ideal fluids".

$$\boxed{P_{ij} = \underset{\substack{\uparrow \\ \text{pressure}}}{p}\,\delta_{ij} \; : \; \text{IDEAL FLUIDS "DRY WATER"}}$$

With this simplification, we can now write the equation of motion as:

← because pressure acts in direction opposite to $d\vec{A}$

$$\int_V \frac{d(\vec{v})}{dt}\rho\, dV = \int_V \rho\vec{g}\, dV - \oint_A p\, d\vec{A}$$

Recalling that $\frac{d\vec{v}}{dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\vec{\nabla})\vec{v}$ and using Gauss-theorem for the pressure term

$$\rightarrow \int_V \rho\frac{d\vec{v}}{dt}\, dV = \int_V \rho\vec{g}\, dV - \int_V dV\vec{\nabla} p - \int_V \rho(\vec{v}\cdot\vec{\nabla})\vec{v}$$

Since this has to hold for any volume V, we can write

$$\boxed{\frac{\partial \vec{v}}{\partial t} = -\frac{1}{\rho}\vec{\nabla} p + \vec{g} - (\vec{v}\cdot\vec{\nabla})\vec{v}}$$

FLUID equation of motion for ideal fluid or EULER EQUATION

The generalization of this equation to include viscosity (and hence shear stresses) is called the Navier-Stokes equ. This equation becomes the Euler equation when $\eta_{visc} = 0$. In term of the Lagrangian derivative, the Euler equation is written as:

$$\boxed{\frac{d\vec{v}}{dt} = \vec{g} - \frac{\vec{\nabla} p}{\rho}}$$

EULER EQUATION

Combining the Euler equation with the continuity equation, one can easily obtain the equation for the momentum, $\rho\vec{v}$:

$$\frac{d(\rho\vec{v})}{dt} = \rho\frac{d\vec{v}}{dt} + \vec{v}\frac{d\rho}{dt} = \rho\vec{g} - \vec{\nabla} p + \vec{v}(-\rho\vec{\nabla}\cdot\vec{v})$$

$$\Rightarrow \boxed{\frac{d(\rho\vec{v})}{dt} = \rho\vec{g} - \vec{\nabla} p - \rho\vec{v}(\vec{\nabla}\cdot\vec{v})}$$

MOMENTUM EQUATION

Note that this equation is dependent on the other two, therefore it can be used in place of the Euler equation, if more convenient in the problem under consideration.

To make this equation more intuitive, note that $\frac{d(\rho\vec{v})}{dt} = \frac{\partial(\rho\vec{v})}{\partial t} + \vec{v}\cdot\vec{\nabla}(\rho\vec{v})$ and the identity:

$\rho v_i(\vec{\nabla}\cdot\vec{v}) + \vec{v}\cdot\vec{\nabla}(\rho v_i) = \vec{\nabla}\cdot(\rho v_i\vec{v})$. Using both of these

we can write: $\frac{d(\rho v)_i}{dt} + \rho v_i(\vec{\nabla}\cdot\vec{v}) = \frac{\partial}{\partial t}(\rho v_i) + \vec{v}\cdot\vec{\nabla}(\rho v_i) + \rho v_i(\vec{\nabla}\cdot\vec{v})$

$= \frac{\partial}{\partial t}(\rho v_i) + \vec{\nabla}\cdot(\rho v_i \vec{v})$. Plugging back in the momentum equation, we obtain:

$$\frac{\partial}{\partial t}(\rho v_i) = \rho g_i - (\vec{\nabla}p)_i - \vec{\nabla}\cdot(\rho v_i \vec{v})$$

Integrating over volume and converting the last integral to a surface integral through Gauss's theorem, one can write

$$\int_V \frac{\partial}{\partial t}(\rho\vec{v}) = \int_V \rho\vec{g}\, dV - \int_V \vec{\nabla}p\, dV - \oint_A \rho\vec{v}(\vec{v}\cdot d\vec{A})$$

The last term can now be recognized as representing the outward flux of momentum [compare with analogous term for ρ: $\oint_A \rho(\vec{v}\cdot d\vec{A})$]

3. The equation of energy

This equation follows from the first law of thermodynamics, which is essentially a statement of the conservation of energy: $dQ = dU + p\,dV$; this law says that if we have a fluid in thermodynamic equilibrium to which some amount of heat has been added and which does the work $p\,dV$ (work done by the system), then the internal energy of the system varies by $dU = dQ - p\,dV$. Before continuing, note that all thermodynamic variables can broadly be classified into two categories: extensive and intensive. The former are variables which are added up when we combine several thermodynamic systems.

An example is the internal energy, or the heat. If we combine two systems with internal energies U_1 and U_2, the internal energy of the resultant system is $U_1 + U_2$. On the other hand, variables like pressure and temperature cannot be added like this. They are called intensive variables. Now, in order to derive an energy equation that does not depend on the amount of fluid that we are considering, we should derive it by defining the extensive properties per unit mass. The energy equation then becomes: $dq = d\epsilon + p\,d\left(\frac{1}{\rho}\right)$ having used the fact that $dV = \frac{dM}{\rho}$ $\left[dM = \rho\, dV \rightarrow dV = \frac{1}{\rho} dM \rightarrow \frac{dV}{dM} = \underset{\substack{\uparrow \\ \text{volume} \\ \text{per unit mass}}}{dv} = \frac{1}{\rho} \right]$

Dividing by dt: $\frac{dq}{dt} = \frac{d\epsilon}{dt} + p\frac{d}{dt}\left(\frac{1}{\rho}\right)$

Note that $p\frac{d}{dt}\left(\frac{1}{\rho}\right) = -\frac{p}{\rho^2}\underbrace{\frac{d\rho}{dt}}_{-\rho\vec{\nabla}\cdot\vec{v} \text{ from the cont. equ.}} = +\frac{p}{\rho}\vec{\nabla}\cdot\vec{v}$

then: $\frac{dq}{dt} = \frac{d\epsilon}{dt} + \frac{p}{\rho}\vec{\nabla}\cdot\vec{v} \Rightarrow \rho\frac{d\epsilon}{dt} + p\vec{\nabla}\cdot\vec{v} = \rho\frac{dq}{dt}$

Now, note that $\frac{dq}{dt}$ is the heat gain (if $dq>0$) or loss (for $dq<0$) per unit mass.

Therefore $\rho\frac{dq}{dt} = (\text{mass} \times \text{unit volume}) \times (\text{heat gain} \times \text{unit mass})$
$= \text{heat gain} \times \text{unit volume} \equiv -\underset{\substack{\uparrow \\ \text{heat loss rate}}}{\mathcal{L}}$

Let us now write $\mathcal{L}$ in terms of our thermodynamic variables

First, note that the heat flux $\vec{\mathcal{F}}$ inside a continuous substance can generally be assumed to be proportional to the temperature gradient, $\vec{\mathcal{F}} = -k\vec{\nabla}T$, where k is called the <u>coefficient of thermal conductivity</u>, and the

negative sign implies that the heat flow is in the direction of decreasing temperature. The heat loss rate from a volume of fluid is equal to the heat flux integrated over the bounding surface of the volume, which is

$$\oint_A \vec{F}\cdot d\vec{S} \underset{\substack{\uparrow \\ \text{Gauss's theorem}}}{=} \int_V \vec{\nabla}\cdot\vec{F}\, dV \quad \Rightarrow \vec{\nabla}\cdot\vec{F} = \text{heat loss rate per unit volume} \equiv \mathcal{L}$$

$$\Rightarrow \mathcal{L} = \vec{\nabla}\cdot\vec{F} = -\vec{\nabla}\cdot(k\vec{\nabla}T)$$

Substituting in the energy equation, and explicitly writing out the Lagrangian derivative, we get:

ENERGY EQUATION

$$\boxed{\rho\left(\frac{\partial \epsilon}{\partial t} + \vec{v}\cdot\vec{\nabla}\epsilon\right) - \vec{\nabla}\cdot(k\vec{\nabla}T) + p\vec{\nabla}\cdot\vec{v} = 0}$$

or

$$\rho\frac{d\epsilon}{dt} = \underbrace{\vec{\nabla}\cdot(k\vec{\nabla}T)}_{\text{heat gain}} \underbrace{-p\vec{\nabla}\cdot\vec{v}}_{\substack{\text{work done} \\ \text{on the system} \\ \text{(when '−' sign in} \\ \text{front)}}}$$

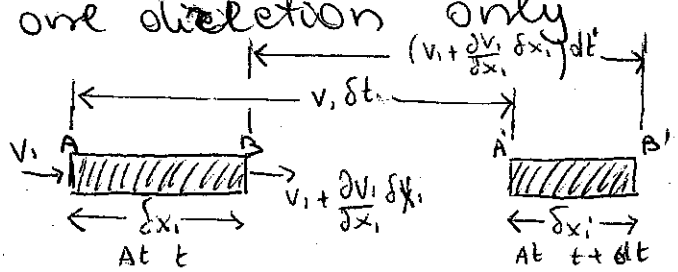
VISCOUS FLOWS & KINEMATICS OF MOTION

VISC

Up to this moment we have considered fluids under very idealized conditions, namely we have neglected the very important role of viscosity. Mathematically, this meant that we assumed all the forces acting on an element of fluid to be perpendicular to its surface at any point of the surface. This is not the behavior of real fluids, in which shear forces act tangentially between moving elements of fluid.

In order to generalize the expression for the forces acting on a fluid element, we first need to study in detail the amount and nature of its deformation (or strain) during its motion. This is because the forces (or stresses) depend on the strains. Note that the deformation of a fluid is similar to that of a solid, where one defines "normal strain" as the change in length per unit length of a linear element, and "shear strain" as a change of a 90° angle. Analogous quantities are defined in a fluid flow, the basic difference being that one defines strain "rates" in a fluid because it continues to deform.

Let's first analize the case in which the deformation occurs in one direction only



A'B' = AB + BB' − AA'

LINEAR or NORMAL STRAIN RATE

Let's consider a fluid element of length $AB = \delta x_1$ at a time t. Let the velocity of point A to be $V_A = V_1$, and the velocity of point B

to be $V_B = V_1 + \frac{\partial V_1}{\partial x_1}\delta x_1$. After a time interval dt, the element of fluid will have a new length given by $A'B' = AB + BB' - AA'$, where $AA' = V_1 \delta t$ is the length traveled by A, and $BB' = \left(V_1 + \frac{\partial V_1}{\partial x_1}\delta x_1\right)dt$ is the length traveled by B.

The linear strain rate is defined as the rate of change of length per unit length, i.e. $\frac{1}{\delta x_1}\frac{d}{dt}(\delta x_1) = \frac{1}{dt}\frac{A'B' - AB}{AB}$

$$= \frac{d}{dt}\frac{1}{\delta x_1}\left[\left(V_1 + \frac{\partial V_1}{\partial x_1}\delta x_1\right)dt - V_1 dt\right] = \frac{\partial V_1}{\partial x_1}$$

Lagrangian derivative since we have followed a fluid element.

In general: LINEAR STRAIN IN i DIRECTION $\boxed{\frac{\partial V_i}{\partial x_i}}$ (no summation implied here)

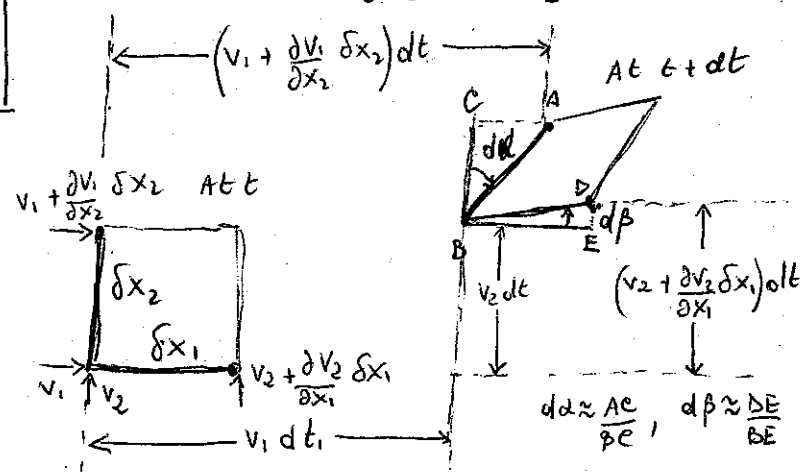
the sum of the linear strain rates in the three mutually orthogonal directions gives the rate of change of volume per unit volume, or volumetric strain rate: $\frac{1}{\delta V}\frac{d}{dt}(\delta V) = \frac{1}{\delta x_1 \delta x_2 \delta x_3}\frac{d}{dt}(\delta x_1 \delta x_2 \delta x$

$$= \frac{1}{\delta x_1}\frac{d}{dt}(\delta x_1) + \frac{1}{\delta x_2}\frac{d}{dt}(\delta x_2) + \frac{1}{\delta x_3}\frac{d}{dt}(\delta x_3) = \frac{\partial V_1}{\partial x_1} + \frac{\partial V_2}{\partial x_2} + \frac{\partial V_3}{\partial x_3}$$

$$\Rightarrow \boxed{\frac{1}{\delta V}\frac{d}{dt}(\delta V) = \sum_i \frac{\partial V_i}{\partial x_i}}$$

SHEAR STRAIN RATE.

In addition to undergoing normal strain rates, a fluid element may also simply deform in shape.



The shear strain rate of an element is defined as the rate of decrease of the angle formed by two mutually perpendicular lines on the element. The shear rate so calculated depends on the orientation of the line pair. The figure shows the position of an element with sides parallel to the coordinate axes at time t, and its subsequent position

at time $t+\delta t$. The rate of shear strain is $\frac{d\alpha+d\beta}{dt}$. From the figure, we see that $d\alpha \approx \frac{AC}{BC}$ (for small angles $tg(d\alpha) \approx d\alpha$) and $d\beta \approx \frac{DE}{BE}$,

where $AC = \left(V_1 + \frac{\partial V_1}{\partial x_2}\delta x_2\right)dt - V_1\,dt \approx \frac{\partial V_1}{\partial x_2}\delta x_2\,dt$, $BC = \delta x_2$,

$BE = \delta x_1$ and $DE = \left(V_2 + \frac{\partial V_2}{\partial x_1}\delta x_1\right)dt - V_2\,dt = \frac{\partial V_2}{\partial x_1}\delta x_1\,dt$

$$\Rightarrow \frac{d\alpha+d\beta}{dt} = \frac{d}{dt}\left\{\frac{1}{\delta x_2}\left(\frac{\partial V_1}{\partial x_2}\delta x_2\,dt\right) + \frac{1}{\delta x_1}\left(\frac{\partial V_2}{\partial x_1}\delta x_1\right)dt\right\} = \frac{\partial V_1}{\partial x_2} + \frac{\partial V_2}{\partial x_1}$$

It is customary to define the

STRAIN RATE TENSOR

$$\boxed{\Lambda_{ij} = \frac{1}{2}\left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i}\right)}$$

This is a symmetric tensor whose diagonal terms are the normal strain rates and whose off diagonal terms are half the shear strain rates.

$$\Lambda_{ij} = \begin{pmatrix} \frac{\partial V_1}{\partial x_1} & \frac{1}{2}\left(\frac{\partial V_1}{\partial x_2} + \frac{\partial V_2}{\partial x_1}\right) & \frac{1}{2}\left(\frac{\partial V_1}{\partial x_3} + \frac{\partial V_3}{\partial x_1}\right) \\ (\ ) & \frac{\partial V_2}{\partial x_2} & \frac{1}{2}\left(\frac{\partial V_2}{\partial x_3} + \frac{\partial V_3}{\partial x_2}\right) \\ (\ ) & (\ ) & \frac{\partial V_3}{\partial x_3} \end{pmatrix}.$$

Given this general expression for the strain, one can then demonstrate that any velocity gradient can be written as the sum of a shear and a rotation. In order to demonstrate this, we need to derive a general expression for the rotation in the same way we did for the shear. To define ROTATION unambiguosly, two mutually perpendicular lines are taken, and the average rotation rate of the two lines is calculated.

Consider the two perpendicular line elements BC and BE in the previous figure. The angular velocities of the line elements about the x_3 axis are $\frac{d\beta}{dt}$ for BE and $-\frac{d\alpha}{dt}$ for BC. So the average is $\frac{1}{2}\left(\frac{d\beta}{dt} - \frac{d\alpha}{dt}\right) =$

$$= \frac{1}{2}\frac{1}{dt}\left\{\frac{1}{\delta x_1}\left(\frac{\partial V_2}{\partial x_1}\delta x_1\, dt\right) - \frac{1}{\delta x_2}\left(\frac{\partial V_1}{\partial x_2}\delta x_2\, dt\right)\right\} = \frac{1}{2}\left(\frac{\partial V_2}{\partial x_1} - \frac{\partial V_1}{\partial x_2}\right).$$

Note that the angular velocity around the axis x_3, turns out to be half the x_3 component of the vorticity: $\vec{\Omega} = \vec{\nabla}\times\vec{V}$, $\vec{\omega} = \frac{1}{2}\vec{\Omega}$

angular velocity of fluid element — vorticity of fluid element

Now we can easily see that any velocity gradient can be written as

$$\frac{\partial V_i}{\partial x_j} = \frac{1}{2}\left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial V_i}{\partial x_j} - \frac{\partial V_j}{\partial x_i}\right) = \Lambda_{ij} + R_{ij}$$

where we have defined the rotation tensor

$$R = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$$

$R_{ij} = -\varepsilon_{ijk}\,\omega_k$

ex: $R_{12} = -\varepsilon_{12k}\,\omega_k = -\varepsilon_{123}\,\omega_3 = -\omega_3$

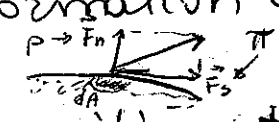
Let's now go back to the momentum equation, which we wrote as $\int_V \frac{dV_i}{dt}\rho\, dV = \int_V \rho g_i\, dV - \underbrace{\oint_A P_{ij}\, dS_j}_{\int_V \frac{\partial P_{ij}}{\partial x_j} dV \text{ (by Gauss's Theorem)}}$

The momentum equation for a small fluid element of volume dV is then (for the i component):

$$\frac{dV_i}{dt}\rho = \rho g_i - \frac{\partial P_{ij}}{\partial x_j}.$$

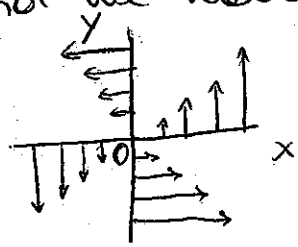
Here $P_{ij} \equiv$ stress tensor (force/area in direction i due to stress in direction j).

Up to this moment we have limited ourselves only to the case in which P_{ij} is a diagonal tensor (i.e. with forces always $\perp$ surface). Now we consider the general case where tangential forces can be present, and we write $P_{ij} = p\delta_{ij} + \Pi_{ij}$, where Π_{ij} is a non-diagonal tensor which contains the information on the shear forces acting on the fluid.



From a qualitative point of view, it is intuitive that the shear force acting between two fluid elements will be proportional to the velocity gradient between those elements

Newton went beyond this by postulating that the shear force is proportional to the velocity gradient: $\Pi_{ij} = -\mu \frac{dv_i}{dx_j}$, where μ is the coefficient of viscosity. Fluids obeying the proportionality relation between the shear stress and the velocity gradient are known as <u>Newtonian fluids</u>. It is found experimentally that most fluids are indeed very close to being Newtonian. For the rest of this course we will limit ourselves to this type of fluids, and we have to find a general expression for Π_{ij}.



The following simple consideration shows that the expression $\Pi_{ij} = -\mu \frac{dv_i}{dx_j}$ could not be a general expression for the shear stress. Suppose a fluid mass is undergoing rigid rotation around the z-axis passing through the point O. Let's look at the velocity gradients $\frac{\partial v_x}{\partial y}$ and $\frac{\partial v_y}{\partial x}$. Certainly they are non-zero, since, for the fluid to be in a solid rotation, the outer layers have to move at increasingly larger speeds compared with the inner layers. However, the solid rotation means that there are no relative motions inside the fluid, and as such, shear forces should be zero. In order to have an expression for the shear stress that depends only on the relative shear motion between two adjacent elements of fluid, we need to substract out the rotation component from dv_i/dx_j. By recalling that $\frac{\partial v_i}{\partial x_j} = \Lambda_{ij} + R_{ij}$, it is immediately derived that we need to impose $\Pi_{ij} \propto \Lambda_{ij}$ (stress $\propto$ strain).

For a Newtonian fluid, the shear stress depends linearly on the velocity gradient. The relation $\Pi_{ij} \propto \Lambda_{ij}$ is however not the most general expression of a tensor Π_{ij} depending linearly

on the velocity gradient. It is a result of tensor analysis that the most general second-rank tensor linearly depending on symmetric combinations of velocity gradients is of the form

$$\boxed{\pi_{ij} = -\left[\eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + \eta' \delta_{ij} \vec{\nabla}\cdot\vec{v}\right]}$$

SHEAR STRESS TENSI

– sign simply as a result of our definition $\pi_{ij} = -dv_i/dx_j$

[note that the above expression can also be derived directly from microscopic considerations].

The additional term adds $\eta' \vec{\nabla}\cdot\vec{v}$ to all diagonal elements of the stress tensor. If the liquid is incompressible, $\vec{\nabla}\cdot\vec{v}=0$ and this extra term does not appear. So it has to do with internal forces generated during compression. The coefficient η is called "first coefficient of viscosity" or "shear viscosity coefficient", while the new coefficient η' is called the "second coefficient of viscosity".

With the expression above for π_{ij}, the general expression for the stress tensor becomes

$$P_{ij} = +p\delta_{ij} - \left[\underbrace{\eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)}_{2\Lambda_{ij}} + \eta' \delta_{ij} \vec{\nabla}\cdot\vec{v}\right] = p\delta_{ij} - [2\eta\Lambda_{ij} + \eta'\delta_{ij}\vec{\nabla}\cdot\vec{v}]$$

$f_i = -\partial P_{ij}/\partial x_j$

Now, the diagonal terms of Λ_{ij} in a flow may be unequal. In such a case the stress tensor P_{ij} can have unequal diagonal terms. We can therefore take the average of the diagonal terms of P_{ij} and define a <u>mean pressure</u> $\bar{P} \equiv +\frac{1}{3}\underbrace{\sum_i P_{ii}}_{\text{Trace (i.e. sum of diagonal element}}$ (as opposed to thermodynamic pressure p).

Note that we can write $\vec{\nabla}\cdot\vec{v} = \text{trace}(\Lambda_{ij}) = \sum_m \Lambda_{mm}$

Since $\sum_i \delta_{ii} = 3$, we get: $\text{Trace}(P_{ij}) = 3p - [2\eta\,\vec{\nabla}\cdot\vec{v} + 3\eta'\vec{\nabla}\cdot\vec{v}]$

$\Rightarrow \bar{P} = p - \left(\frac{2}{3}\eta + \eta'\right)\vec{\nabla}\cdot\vec{v} \Rightarrow p - \bar{P} = \left(\frac{2}{3}\eta + \eta'\right)\vec{\nabla}\cdot\vec{v}$. Now, for a completely incompressible fluid, we can only define a

mechanical or mean pressure, since there is no equation of state to determine a thermodynamic pressure. For a compressible fluid, on the other hand, a thermodynamic pressure can be defined, and it seems from the previous equation that p and $\bar{p}$ can be different. For most studies of fluids, it is found that p and $\bar{p}$ are equal to a very good approximation. The condition $\frac{2}{3}\eta + \eta' = 0$, called the "Stokes assumption", is also supported from the kinetic theory of monoatomic gases. Given this condition, and the original definition of $\Pi_{ij} = -\mu\, dv_i/dx_j$, we can write

$$\boxed{\Pi_{ij} = -\mu\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\,\vec{\nabla}\cdot\vec{v}\right)}$$

On substituting into the equation of motion $\rho\frac{dv_i}{dt} = \rho g_i - \frac{\partial P_{ij}}{\partial x_j}$ and assuming $\mu = \text{const}$ within the fluid (which is true for most situations) we obtain:

$$\rho\frac{dv_i}{dt} = \rho g_i - \frac{\partial P}{\partial x_i} + \mu\frac{\partial}{\partial x_j}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\,\vec{\nabla}\cdot\vec{v}\right).$$

This is the i component of the Navier-Stokes equation. To write it in vector form, note that

$$\frac{\partial v_i}{\partial x_j \partial x_j} = \frac{\partial v_i^2}{\partial x_1^2} + \frac{\partial v_i^2}{\partial x_2^2} + \frac{\partial v_i^2}{\partial x_3^2} \equiv \nabla^2 v_i$$

and that $\frac{\partial}{\partial x_j}\delta_{ij}\,\vec{\nabla}\cdot\vec{v} = \frac{\partial}{\partial x_i}\vec{\nabla}\cdot\vec{v}$

also: $\frac{\partial}{\partial x_j}\frac{\partial v_j}{\partial x_i} = \frac{\partial}{\partial x_i}\frac{\partial v_j}{\partial x_j} = \frac{\partial}{\partial x_i}(\vec{\nabla}\cdot\vec{v})$

Therefore, we obtain, in vector form:

NAVIER-STOKES EQUATION

$$\boxed{\rho\frac{d\vec{v}}{dt} = \rho\vec{g} - \vec{\nabla}P + \mu\left[\nabla^2\vec{v} + \frac{1}{3}\vec{\nabla}(\vec{\nabla}\cdot\vec{v})\right]}$$

The term involving $\vec{\nabla}(\vec{\nabla}\cdot\vec{v})$ has significance only in the case of flows with variable compression. An example is the dissipation of acoustic waves. However, in the greatest majority of cases, it can be neglected and the Navier-Stokes equation reduces to:

$$\boxed{\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\vec{\nabla})\vec{v} = \vec{g} - \frac{1}{\rho}\vec{\nabla}P + \nu\nabla^2\vec{v}}$$

where $\nu = \mu/\rho$ is called kinematic viscosity.

Note that the last version of the Navier-Stokes equation reduces to the Euler equation for $\nu = 0$. The mathematical characters of the two equations, however, are vastly different due to the fact that the Navier-Stokes equation contains a higher spatial derivative than the Euler equation. Hence the Navier-Stokes equation requires more boundary conditions for solutions in finite regions. In particular: while solving ideal fluid problems with solid boundaries, one usually takes the normal component of the velocity to be zero on the solid surface. On the other hand, while solving viscous fluid problems, one imposes the extra boundary condition on a solid surface that the tangential component of velocity is also zero making $\vec{v} = 0$ there. Observations from everyday life convince us that this is actually the correct boundary condition for fluid flows along solid boundaries. For example, we notice that the blades of a fan collect dust. Even though the fan may be in regular use to produce winds, the air velocity at the surface of the blades remains zero so that the dust can collect there. Similarly, if one tries to sweep away fine dust particles from a solid surface by blowing air on it, one will realize how difficult it is. No matter how hard one blows, the air velocity exactly at the surface remains zero.

As a last consideration on the Navier-Stokes equation, let's take the curl of it and recall that $\vec{\nabla} \times \vec{v} = \vec{\Omega}$ (vorticity). The only term that differs from before (i.e. when we did the same thing with the Euler equation) is the last one. Using the

identity $\nabla \times \nabla^2 V = \nabla^2 (\nabla \times V) = \nabla^2 \Omega$, we obtain:

$$\boxed{\frac{\partial \vec{\Omega}}{\partial t} = \vec{\nabla} \times (\vec{V} \times \vec{\Omega}) + \nu \nabla^2 \vec{\Omega}}$$ for incompressible fluids.

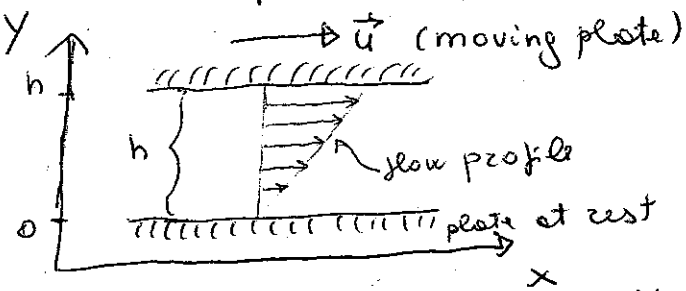
The main difference with the no viscosity case (ideal fluids) is the presence of the term $\nu \nabla^2 \vec{\Omega}$. Because of this term, Kelvin's vorticity theorem no longer holds for viscous fluids, making the production and decay of vorticity possible. The complete dynamical theory of viscous, incompressible fluids, is provided by the above equation in combination with $\vec{\nabla} \cdot \vec{V} = 0$ and $\vec{\Omega} = \vec{\nabla} \times \vec{V}$. Also for viscous fluids the energy equation in redundant when one considers incompressible fluids, otherwise it is needed.

For non-viscous fluids, we derived the energy equation from the first law of thermodynamics: $\rho \frac{d\epsilon}{dt} = \vec{\nabla} \cdot (K \vec{\nabla} T) - p \vec{\nabla} \cdot \vec{V}$.

If there is dissipation, the internal energy will increase also as a result of conversion of mechanic energy into it. If we indicate this term by ϕ_{visc}, the energy equation generalizes to:

$$\boxed{\underbrace{\rho \frac{d\epsilon}{dt}}_{\text{increase in internal energy}} = \underbrace{\vec{\nabla} \cdot (K \vec{\nabla} T)}_{\text{flux of heat into region}} \underbrace{- p \vec{\nabla} \cdot \vec{V}}_{\text{compression on system}} + \underbrace{\phi_{visc}}_{\text{heating due to viscous dissipation.}}}$$ Energy Equation

Example 1: Flow between plates.



A flow is in between two plates, one at rest, and the other moving with velocity $\vec{u} = u\hat{e}_x$.

Moreover;
- pressure gradient along $x = 0$
- $\vec{v} = v(y)\,\hat{e}_x$ (for an incompressible flu the velocity of the fluid will not depend on $\hat{e}_x$) from continuity equation

* flow is produced by motion of the upper plate, and not by pressure gradient

* Find $v(y)$

$\vec{\nabla}\cdot\vec{v} = 0 \rightarrow \frac{\partial v}{\partial x} = 0 \rightarrow v$ indep. of $x \rightarrow \vec{v} = v(y)\,\hat{e}_x$

The velocity profile is determined by solving the Navier-Stokes equations: $\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\vec{\nabla})\vec{v} = -\frac{1}{\rho}\vec{\nabla}P + \nu\nabla^2\vec{v}$

In steady conditions: $\frac{\partial \vec{v}}{\partial t} = 0$

The advection term is $(\vec{v}\cdot\vec{\nabla})\vec{v} = (v(y)\,\hat{e}_x\cdot\vec{\nabla})\,v(y)\,\hat{e}_x$
$= v(y)\frac{\partial}{\partial x}v(y) = 0$

So: $\frac{1}{\cancel{\rho}}\vec{\nabla}P = \frac{\mu}{\cancel{\rho}}\nabla^2\vec{v} \Rightarrow$ x-comp. $0 = \frac{d^2v}{dy^2} \Rightarrow v(y) = ay + b$

y-comp $\frac{dp}{dy} = 0 \Rightarrow p(y) =$ const

The velocity profile is linear, $v(y) = ay + b$, and the parameters a and b are determined from the boundary conditions:

(no-slip boundary conditions; i.e. velocity of flow at plate = velocity of plate)

$$\begin{cases} v(0) = 0 \Rightarrow b = 0 \\ v(h) = u \Rightarrow a = \frac{u}{h} \end{cases} \Rightarrow \boxed{v(y) = \left(\frac{u}{h}\right)y}$$

Mean flow velocity: $\bar{v} = \frac{1}{h}\int_0^h v(y)\,dy = \frac{u}{2}$

Force on plane $(y = h)$

$$\Pi_{xy} = -\mu\frac{dv}{dy} = -\frac{\mu u}{h}$$

$$\frac{\text{Force}}{\text{Area}} = \Pi_{xy} = -\frac{\mu u}{h}$$